1. Preliminaries

Definition 1.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

Definition 1.2 (Central factorials). *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n\left(n + \frac{k}{2} - 1\right)\left(n + \frac{k}{2} - 2\right)\cdots\left(n - \frac{k}{2} + 1\right) = n\prod_{j=1}^{k-1}\left(n + \frac{k}{2} - j\right), & \text{if } k > 0. \end{cases}$$

Proposition 1.3 (Central Newton's formula). *For arbitrary integers x, and t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes central finite difference of $f(t)$, defined by

$$\delta^k f(t) = \sum_{j=0}^{k} (-1)^j \tbinom{k}{j} f(t + \tfrac{k}{2} - j).$$

Proof. See [**?**, p. 462]. □

Lemma 1.4 (Central falling factorials). *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n\left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0. \end{cases}$$

where $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1}\left(n + \frac{k}{2} - j\right)$ are falling factorials.

Lemma 1.5 (Binomial form of central factorials). *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k}\binom{n + \frac{k}{2} - 1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!}\left(n+\frac{k}{2}-1\right)_{k-1} = \frac{n}{k(k-1)!}\left(n+\frac{k}{2}-1\right)_{k-1} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$, and Lemma (1.4). ☐

Lemma 1.6 (Halved central factorials)**.** *For integers $n, k \geq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{1}{2}\binom{n+\frac{k}{2}}{k} + \frac{1}{2}\binom{n+\frac{k}{2}-1}{k}.$$

Proof. We have $\binom{a}{k} = \frac{a}{k}\binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\tbinom{n+\frac{k}{2}}{k} = \frac{n+\frac{k}{2}}{k}\tbinom{n+\frac{k}{2}-1}{k-1} = \tfrac{n}{k}\tbinom{n+\frac{k}{2}-1}{k-1} + \tfrac{1}{2}\tbinom{n+\frac{k}{2}-1}{k-1}.$$

Thus,

$$\tfrac{n}{k}\tbinom{n+\frac{k}{2}-1}{k-1} = \tbinom{n+\frac{k}{2}}{k} - \tfrac{1}{2}\tbinom{n+\frac{k}{2}-1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\tfrac{n}{k}\tbinom{n+\frac{k}{2}-1}{k-1} = \tfrac{1}{2}\left[2\tbinom{n+\frac{k}{2}}{k} - \tbinom{n+\frac{k}{2}-1}{k-1}\right] = \tfrac{1}{2}\left[\tbinom{n+\frac{k}{2}}{k} + \tbinom{n+\frac{k}{2}-1}{k} + \tbinom{n+\frac{k}{2}-1}{k-1} - \tbinom{n+\frac{k}{2}-1}{k-1}\right].$$

Thus, the statement follows. ☐

Proposition 1.7 (Central Newton's formula for powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \sum_{k=0}^{m} \frac{(n-t)^{[k]}}{k!}\delta^k t^m.$$

2. Mathematica

Mathematica Function	**Validates / Prints**
`MultifoldSumOfPowersRecurrence[r, n, m]`	Computes $\Sigma^r n^m$, (1.1)
`CentralFactorial[n, k]`	Computes $n^{[k]}$, (1.2)
`ValidateCentralFactorialsInTermsOfFalling[10]`	Validates Lem. (1.4)
`ValidateBinomialFormOfCentralFactorials[10]`	Validates Lem. (1.5)
`ValidateHalvedCentralBinomials[20]`	Validates Lem. (1.6)
`ValidateNewtonsFormulaForPowers[10]`	Validates Prop. (1.7)